NETAJI SUBHASH ENGINEERING COLLEGE

SUBJECT :- DATA WAREHOUSING AND DATA MINING (PEC-IT602B)

NAME :- ROHAN KARAK

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SEMESTER :- 6th **STREAM** :- CSE

CLASS ROLL NO.:- 117 **SECTION :-** B

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Q) 1) classification

Assignment 1 :-

Day	Outlook	Temperature	Humidity	wind	Decision; (Golf Play possible)
1	sunny	Hot	High	weak	NO
2	sunny	Hot	High	Strong	NO
3	overcast	Hot	High	weak	yes
4	Rain	Mild	High	Weak	yes
5	Rain	cool	Normal	weak	yes
6	~~overcast~~ Rain	cool	Normal	strong	NO
7	~~sunny~~ overcast	cool	Normal	Strong	yes
8	sunny	Mild	High	weak	NO
9	sunny	cool	Normal	weak	yes
10	Rain	Mild	Normal	weak	yes
11	sunny	Mild	Normal	Strong	yes
12	overcast	Mild	High	strong	yes.
13	overcast	Hot	Normal	weak	yes
14	Rain	Mild	High	strong	NO

Expected information :-

Attribute : Outlook

$info(D) = -\sum_{i=1} p_i \log_2(p_i)$

values(outlook) = sunny, overcast, Rain.

$info(D) = I(9,5)$

$= -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$

$Info_{outlook}(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Info(D_j)$

$= \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2)$

$= \frac{5}{14} \times 0.971 - \frac{4}{14} \times 0 - \frac{5}{14} \times 0.971$

$= 0.694$

$\therefore Gain(D, outlook) = 0.94 - 0.694$

$= 0.2464$

$\because Gain(D, outlook) = info(D) - Info(D)_{outlook}$

$\therefore$ Attribute : Temperature values (Temp) = Hot, Mild, Cool

Similary, $Info(D) = 0.94$

$Info_{temperature}(D) = \frac{4}{14} I(2,2) + \frac{6}{14} I(4,2) + \frac{4}{14} I(3,1)$

$= \frac{4}{14} \times 1 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 0.811$

$= 0.911$

$\therefore Gain(D, Temperature) = 0.94 - 0.911$

$= 0.0289$

$\because$ Attribute : Humidity

values (Humidity) = High, Normal

Similarly, $Info(D) = 0.94$

$Info_{Humidity}(D) = \frac{7}{14} I(3,4) + \frac{7}{14} I(6,1)$

$= \frac{7}{14} \times 0.9852 + \frac{7}{14} \times 0.5916 = 0.788$

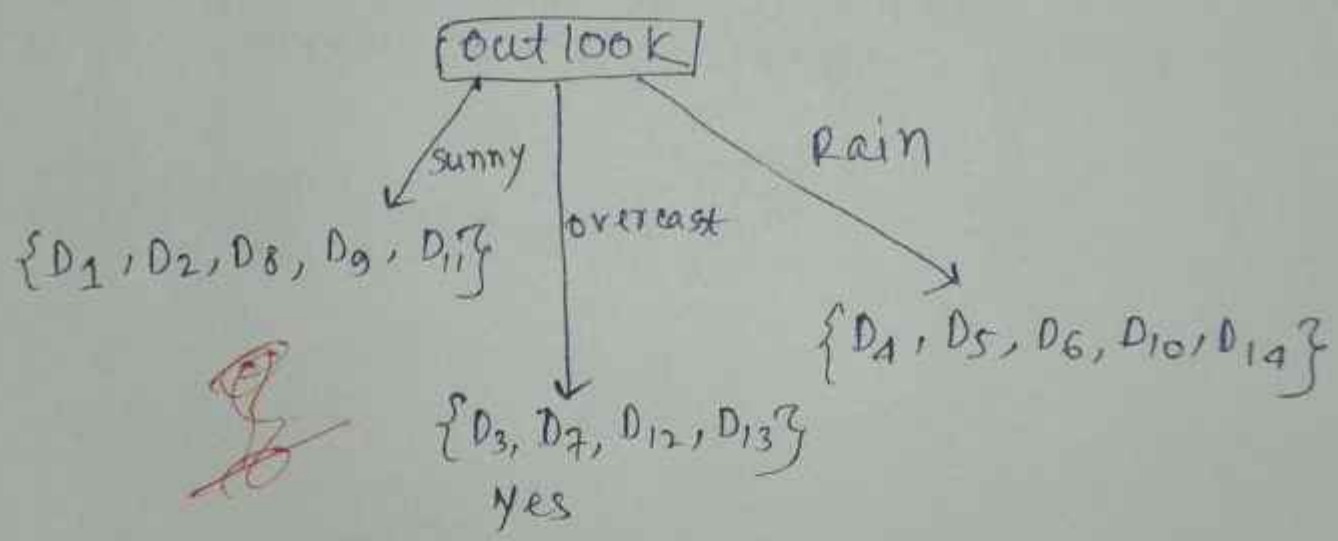
$\therefore$ Gain (D, Humidity) $= 0.94 - 0.788$
$= 0.1516$

Attribute: Wind

similarly, Info(D) = 0.94

$$\text{Info}_{\text{wind}}(D) = \frac{6}{14} \times I(3,3) + \frac{8}{14} \times I(6,2)$$
$$= \frac{6}{14} \times 1 + \frac{8}{14} \times 0.8113$$
$$= 0.892$$

$\therefore$ Gain (D, wind) $= 0.94 - 0.892$
$= 0.0478$



similarly, for D sunny,

Gain (D sunny, Temperature) = 0.570
Gain (D sunny, Humidity) = 0.97
Gain (D sunny, wind) = 0.0192

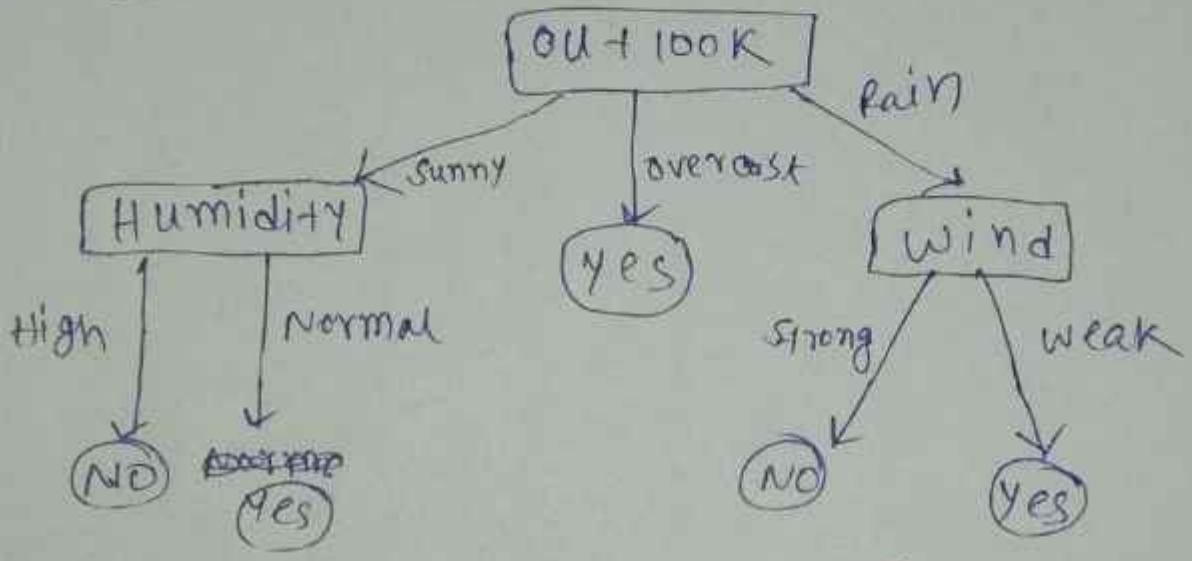
Now, similarly for D_{Rain},

$Gain(D_{Rain}, Temperature) = 0.0192$

$Gain(D_{Rain}, Humidity) = 0.0192$

$Gain(D_{Rain}, Wind) = 0.97$

Decision Table :-



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Date :- 07/02/2026

Assignment - 2

Assignment = II

(a) complete link Hierachial clustering :-
Also draw the dendrogram;

	P_1	P_2	P_3	P_4	P_5
P_1	0.00	0.10	0.41	0.55	0.35
P_2	0.10	0.00	0.64	0.47	0.98
P_3	0.41	0.64	0.00	0.44	0.85
P_4	0.55	0.47	0.44	0.00	0.76
P_5	0.35	0.98	0.87	0.76	0.00

Table-1

∵ complete link = max value to be put in updated matrix after merging;

$D_{complete} = \max_{x,y} \{d(x,y) \mid x \in C_i, y \in C_j\}$

(minimum value = 0.10 {from P_1 & P_2})
thus, merging P_1 & P_2 together (taking maximum value b/w P_1 & P_2 to other points)
let, C_1 = combination of P_1 & P_2;

	C_1	P_3	P_4	P_5
C_1	0.00	0.64	0.55	0.98
P_3	0.64	0.00	0.44	0.85
P_4	0.55	0.44	0.00	0.76
P_5	0.98	0.85	0.76	0.00
~~P_1~~	×	×	×	×

Table-2

(minimum value = 0.44 {from P_3 & P_4})
thus, merging P_3 & P_4 together (taking maximum value b/w P_3 & P_4 to other points)
let C_2 = combination of P_3 & P_4;

	C_1	C_2	P_5
C_1	0.00	0.64	0.48
C_2	0.64	0.00	0.85
P_5	0.98	0.85	0.00

Table - 3

(minimum value = 0.64 {from C_1 & C_2})
thus, merging C_1 & C_2 together (taking maximum values b/w C_1 & C_2 to other points)
let C_3 = C_1 combined with C_2;

	C_3	P_5
C_3	0.00	0.98
P_5	0.98	0.00

Table 4

(minimum value = 0.48 (from C_3 & P_5)) thus, merging C_3 & P_5 together (taking maximum values b/w C_3 & P_5 to complete the loop)

let $C_3 = C_3$ combined with P_5)

Now,

	C_4
C_4	0.00

Table 5

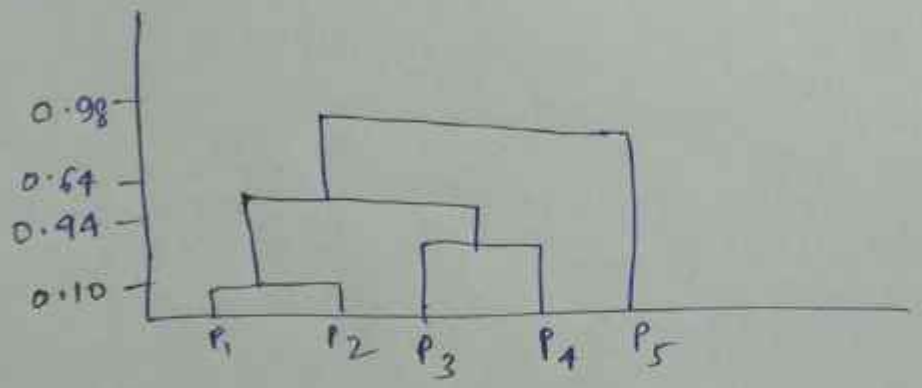
now, $C_4 = C_3 + P_5$
$= (C_1 + C_2) + P_5$
$= ((P_1 + P_2) + (P_3 + P_4)) + P_5$

Thus,

	P_1, P_2, P_3, P_4, P_5
P_1, P_2, P_3, P_4, P_5	0.00

{all iterations of merging completed;}
Sequense of statements is as follows:-
$((P_1, P_2), (P_3, P_4), P_5)$

The dendrogram is as follows:-



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Sub:- DWDM (Assignment III)

Suppose the data mining task is to cluster the following 8 points (with (x, y) representing locations. Suppose 3 clusters are to be formed. A1(1,1), A2(3,3), A3(1,4), A4(5,5), A5(6,6), A6(9,9), A7(0,3), A8(3,0). Euclidean distance is used as the distance function. Initially, A2(3,3), A4(5,5) and A5(6,6) are assigned as the centroid of each cluster. Use the k-means algorithm to determine the 3 clusters.

Ans:- Initial cluster centers are —
A2(3,3), A4(5,5) and A5(6,6)

Iteration 1:-

Data points	Distance to			Cluster	New cluster
	c1:(3,3)	c2:(5,5)	c3:(6,6)		
A1(1,1)	$\sqrt{8}$	$\sqrt{32}$	$\sqrt{50}$	1	—
A2(3,3)	0	$\sqrt{8}$	$\sqrt{18}$	1	—
A3(1,4)	$\sqrt{2}$	$\sqrt{2}$	$\sqrt{8}$	1	—
A4(5,5)	$\sqrt{8}$	0	$\sqrt{2}$	2	—
A5(6,6)	$\sqrt{18}$	$\sqrt{2}$	0	3	—
A6(9,9)	$\sqrt{12}$	$\sqrt{32}$	$\sqrt{18}$	3	—
A7(0,3)	$\sqrt{9}$	$\sqrt{29}$	$\sqrt{45}$	1	—
A8(3,0)	$\sqrt{9}$	$\sqrt{29}$	$\sqrt{45}$	1	—

$$d(i, j) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

New clusters: cluster 1: {A1, A2, A3, A7, A8}
cluster 2: {A4}
cluster 3: {A5, A6}

Re-compute the new cluster centers (means)
for, cluster 1: ((1+3+4+0+3)/5, (1+3+4+3+0)/5)
= (2.2, 2.2)

cluster 2: (5,5), since we only have one data point A4 (5,5)

cluster 3: ((6+9)/2, (6+9)/2)
= (7.5, 7.5)

Iteration 2:-

Data Points	Distance to			cluster (old)	New cluster
	c1: (2.2, 2.2)	c2: (5,5)	c3 (7.5, 7.5)		
A1 (1,1)	$\sqrt{2.88}$	$\sqrt{32}$	$\sqrt{84.5}$	1	1
A2 (5,3)	$\sqrt{1.28}$	$\sqrt{8}$	$\sqrt{40.5}$	1	1
A3 (4,4)	$\sqrt{6.48}$	$\sqrt{2}$	$\sqrt{24.5}$	1	2
A4 (5,5)	$\sqrt{15.68}$	0	$\sqrt{12.5}$	2	2
A5 (6,6)	$\sqrt{28.88}$	$\sqrt{2}$	$\sqrt{4.5}$	3	2
A6 (9,9)	$\sqrt{92.48}$	$\sqrt{32}$	$\sqrt{4.5}$	3	3
A7 (0,3)	$\sqrt{5.48}$	$\sqrt{29}$	$\sqrt{76.5}$	1	1
A8 (3,0)	$\sqrt{5.48}$	$\sqrt{29}$	$\sqrt{76.5}$	1	1

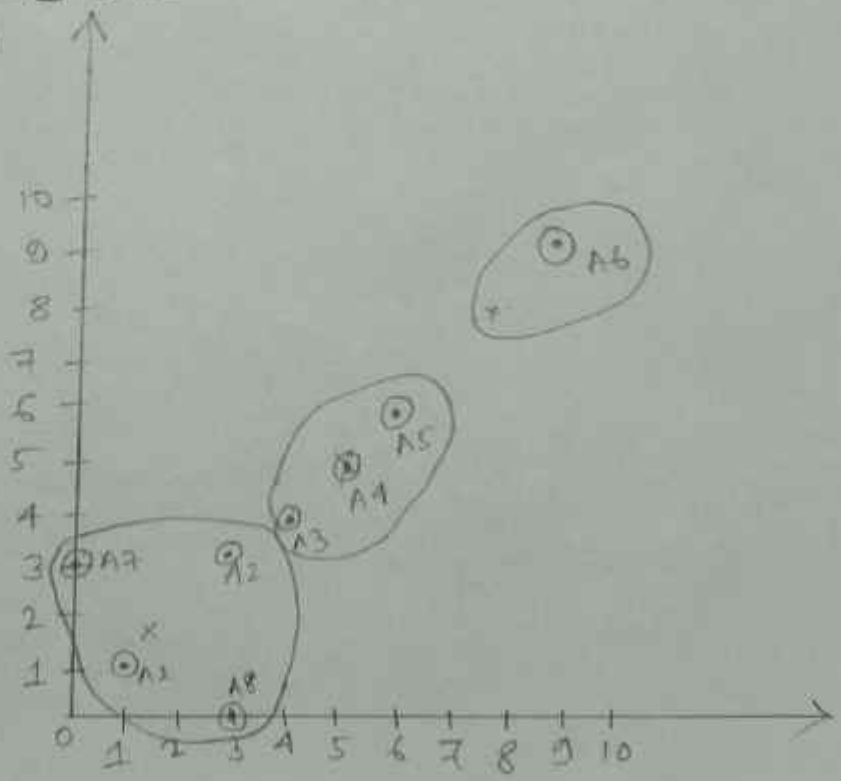
New clusters:- cluster 1: {A1, A2, A7, A8}
cluster 2: {A3, A4, A5}
cluster 3: {A6}

Now, re-compute the new cluster centers
for cluster 1: ((1+3+0+3)/4, (1+3+3+0)/4) = (1.75, 1.75)
cluster 2: ((4+5+6)/3, (4+5+6)/3) = (5,5)
cluster 3: (7.5, 7.5), since we only have one data point.

Iteration :- 3

Data Points	Distance to			Cluster (old)	New cluster
	C1 : (1.75, 1.75)	C2 : (5, 5)	C3 : (7.5, 7.5)		
A1 (1,1)	$\sqrt{1.125}$	$\sqrt{32}$	$\sqrt{84.5}$	1	1
A2 (3,3)	$\sqrt{3.125}$	$\sqrt{8}$	$\sqrt{40.5}$	1	1
A3 (4,4)	$\sqrt{10.125}$	$\sqrt{2}$	$\sqrt{24.5}$	2	2
A4 (5,5)	$\sqrt{21.125}$	0	$\sqrt{12.5}$	2	2
A5 (6,6)	$\sqrt{36.125}$	$\sqrt{2}$	$\sqrt{4.5}$	2	2
A6 (9,9)	$\sqrt{105.13}$	$\sqrt{32}$	$\sqrt{4.5}$	3	3
A7 (0,3)	$\sqrt{4.625}$	$\sqrt{29}$	$\sqrt{76.5}$	1	1
A8 (3,0)	$\sqrt{4.625}$	$\sqrt{29}$	$\sqrt{76.5}$	1	1

New clusters :- Cluster 1 : {A1, A2, A7, A8}
Cluster 2 : {A3, A4, A5}
Cluster 3 : {A6}

Since there is no change in cluster thus we are ending the iteration here.

New cluster centers are —
C1 : (1.75, 1.75)
C2 : (5, 5)
C3 : (7.5, 7.5)